

Cybersecurity Academy — Course Catalog Content (3-Tier Structure)

For: Courses page **Structure:** Fundamentals (Beginner) → CompTIA Security+ (Intermediate) → Specialized Track (choose one: Pentest / SOC / GRC)

Note: This replaces the previous flat 9-track catalog. Former standalone tracks (Linux Administration, Network Security, Cloud Security, Digital Forensics) have been folded in as modules or advanced electives rather than dropped — see notes under each tier. Duration and pricing below are placeholders pending CEO/curriculum sign-off.

Tier 1: Cyber Security Fundamentals

Level: Beginner — *Start Here*

The perfect starting point. Learn the core concepts of information security, threats, vulnerabilities, and defenses.

Duration: 6 Weeks

Price: 200,000.00

Prerequisites: None — ideal for beginners

What You'll Learn: CIA Triad · Network Basics · Threat Actors · Cryptography

Tier 2: Intermediate Tracks

CompTIA Security+ — *Most Popular*

Build toward one of the most recognized entry-level certifications in the industry.

Duration: 12 Weeks

Price: 680,000

Prerequisites: Cyber Security Fundamentals

What You'll Learn: Risk Management · Access Control · Applied Cryptography · Security Architecture **Capstone (per curriculum standard):** Project 1 — Human Firewall / Phishing Awareness Initiative. Project 2 — CYVANT Cybersecurity Innovation Challenge.

Network Security

Understand how data moves — and how attackers intercept, disrupt, or exploit it.

Duration: 8 Weeks

Price: 400,000

Prerequisites: Cyber Security Fundamentals

What You'll Learn: Firewalls & VPNs · Network Protocols · Intrusion Detection · Network Segmentation

Linux Administration

Get comfortable operating in the environment most security tools and servers actually run on.

Duration: 4 Weeks

Price: 250,000

Prerequisites: Cyber Security Fundamentals

What You'll Learn: Command Line Mastery · File Permissions · User Management · Shell Scripting

Active Directory

Learn how most enterprise networks manage identity, access, and authentication — and where that system is commonly attacked.

Duration: 8 Weeks

Price: 400,000

Prerequisites: Cyber Security Fundamentals (Linux Administration recommended alongside)

What You'll Learn: AD Structure & Domains · User & Group Policy Management · Authentication Protocols (Kerberos/NTLM) · AD Attack & Defense Basics

Security Monitoring

Learn the fundamentals of watching a live environment for suspicious activity — the entry point into monitoring-focused security work.

Duration: 4 Weeks

Price: 250,000

Prerequisites: Network Security recommended

What You'll Learn: SIEM Fundamentals · Log Analysis Basics · Alert Triage · Monitoring Tools Overview

Vulnerability Assessment

Learn to systematically identify, score, and report security weaknesses before they're exploited.

Duration: 4 Weeks

Price: 250,000

Prerequisites: Network Security recommended

What You'll Learn: Vulnerability Scanning Fundamentals · CVE & Risk Scoring (CVSS) · Patch Management · Reporting & Remediation Planning

On "Practical Labs": Across all intermediate level courses

Tier 3: Specialized Track (850,000)

Level: Advanced — choose one path

After completing Fundamentals and Security+, students choose a specialization. Each path is built as its own focused track rather than a single generic "advanced" course.

Path A: Penetration Testing (Offensive Security) + Mentorship

Learn to think like an attacker — legally and methodically — to find and report vulnerabilities before someone else exploits them.

Duration: 16 Weeks

Price: 920, 000

Prerequisites: CompTIA Security+ Or equivalent skill level

What You'll Learn: Reconnaissance · Vulnerability Scanning · Exploitation Techniques · Professional Reporting

Advanced elective (optional add-on): Cloud Security — Shared Responsibility Model, Cloud IAM, Cloud Threat Landscape, Securing AWS/Azure Environments

Penetration Testing (Offensive Security) — Full Curriculum Breakdown

Tier: 3A — Advanced Specialization

Aligns with (external certifications, not CYVANT credentials): CEH, CompTIA PenTest+, OSCP-style skill areas

Programme Overview

This track takes students through a complete offensive security engagement — from first recon to final report — using CYVANT's Learn → Practice → Build → Present → Defend → Portfolio → Opportunity approach. Every module pairs core theory with a hands-on lab, and where relevant, an **AI-Assisted Learning** component showing how to use AI tools responsibly to accelerate research, documentation, and analysis (not to replace hands-on skill-building).

Suggested duration: 12 Weeks (*expanded from the current 8-week estimate to properly cover all 12 modules — flag if you'd rather compress this into fewer, denser weeks*)

Module 1: Foundations of Ethical Hacking

- Legal and ethical boundaries of security testing; awareness of major global standards (GDPR, NIST, ISO)
- Understanding hacker classifications: white hat, black hat, gray hat
- The hacking lifecycle: Reconnaissance → Scanning → Gaining Access → Maintaining Access → Covering Tracks
- **AI-Assisted Learning:** using AI tools to break down and summarize ethical considerations and attack-chain logic for study purposes

Module 2: Building Your Hacking Lab

- Installing and navigating a Kali Linux environment
- Core terminal workflows and basic scripting
- Getting oriented in hands-on practice ranges
- **Hands-on Lab:** set up your lab environment and complete your first guided exploit

Module 3: Reconnaissance & Information Gathering

- Passive vs. active reconnaissance techniques
- DNS enumeration, WHOIS lookups, and search-engine-based discovery (e.g. Shodan)
- OSINT tooling, search-engine dorking, and social engineering fundamentals
- **AI-Assisted Learning:** using AI to help structure and summarize footprinting findings into a clean report

Module 4: Scanning & Vulnerability Assessment

- Port and service scanning (Nmap, Masscan)
- OS and service fingerprinting
- Vulnerability scanning (Nessus, OpenVAS, Nikto)
- **Hands-on Lab:** scan a live target machine, interpret the results, and produce a risk summary

Module 5: Exploitation Fundamentals

- Exploiting Windows, Linux, and web-based targets
- Working with the Metasploit Framework
- Introductory brute-force and buffer overflow concepts
- **Toolset covered:** Metasploit, SQLmap, Hydra, John the Ripper

Module 6: Web Application Security

- OWASP Top 10 walkthrough
- SQL injection, XSS, CSRF, IDOR, and file upload vulnerabilities
- Webshells and privilege escalation via web vectors
- **AI-Assisted Learning:** using AI to generate safe, lab-only payload examples for testing understanding
- **Toolset covered:** Burp Suite, OWASP ZAP, Nikto

Module 7: Wireless & Bluetooth Security

- Wi-Fi encryption standards and their weaknesses (WEP, WPA2)
- Packet capture and injection (Aircrack-ng, Wireshark)
- Rogue access points and man-in-the-middle attacks
- **Hands-on Lab:** capture and crack a WPA2 handshake in a fully sandboxed environment

Module 8: Privilege Escalation & Post-Exploitation

- Linux and Windows privilege escalation paths
- Credential harvesting, backdoors, and persistence techniques
- File transfer and network pivoting fundamentals

Module 9: Malware & Social Engineering Awareness

- Malware categories: keyloggers, rootkits, RATs
- Designing authorized phishing simulations for organizational awareness training
- Introductory concepts in obfuscation and evasion
- **AI-Assisted Learning:** using AI to model social engineering risk scenarios for defensive training purposes, always within authorized, ethical, and legal boundaries

Module 10: Evading Security Controls

- Antivirus, EDR, and firewall evasion concepts
- Payload obfuscation fundamentals (msfvenom)
- Overview of AV evasion tooling and how defenders detect it

Module 11: Professional Reporting & Documentation

- Writing a penetration testing report a client can actually act on
- Risk scoring with CVSS; structuring executive vs. technical summaries
- Evidence handling and chain-of-custody basics
- **AI-Assisted Learning:** using AI to help draft and structure report sections, with the student responsible for technical accuracy

Module 12: Capstone Project & Career Readiness

- Simulate a complete engagement end-to-end: recon → exploitation → escalation → documentation → presentation
- Build a portfolio: resume, GitHub with scripts and writeups, and a presentable capstone report
- Exposure to CEH, PenTest+, and OSCP-style challenge formats to prepare for external certification paths

(Per CYVANT's Programme Quality Standard, this capstone should include defined learning outcomes, assessment criteria, a mentor touchpoint, and a public presentation component – consistent with how the Security+ capstone is structured.)

Certification Alignment

This track builds toward (external, third-party certifications):

- CEH (Certified Ethical Hacker)
- CompTIA PenTest+
- OSCP (Offensive Security Certified Professional) – introductory exposure only; full OSCP readiness typically requires additional independent practice beyond this track

Career Paths This Prepares Students For

- Ethical Hacker
- Penetration Tester
- Vulnerability Analyst
- Red Team Specialist
- Security Researcher

CAREER SUPPORT

This track includes structured career support layered on top of the technical curriculum:

- **Portfolio building:** students build 3 high-caliber projects across the track, forming the core of their professional portfolio
- **1 to1 Mentorship & coaching:** delivered per the standalone **CYVANT Mentorship Support Model**, a field-experienced SOC mentor handles office hours, async support, milestone check-ins, and capstone review (see that document for full structure)
- **Study group:** peer accountability structure within the cohort's async support channel
- **Career mapping session:** guidance on how this track connects to specific role targets
- **Job interview preparation**
- **Online presence support:** guidance on building LinkedIn/portfolio visibility for job prospects

● Path B: SOC Operations + Mentorship

Step into the role of a Security Operations Center analyst — monitoring, detecting, and responding to real-time threats.

Duration: 16 Weeks

Price: 850,000

Prerequisites: CompTIA Security+ Or equivalent skill level

What You'll Learn: SIEM Tools · Log Analysis · Incident Triage · Threat Detection **Advanced elective (optional add-on):** Digital Forensics — Evidence Handling, Disk & Memory Forensics, Chain of Custody, Investigation Reporting

SOC Operations — Full Curriculum Breakdown

Tier: 3B — Advanced Specialization **Prerequisites:** CompTIA Security+ (Security Monitoring recommended) **Builds toward (external certifications, not CYVANT credentials):** CompTIA CySA+, Blue Team Level 1, and SIEM-platform-specific certifications (e.g. Splunk Core Certified User) depending on tooling covered

Programme Overview

This track prepares students to work as a Security Operations Center analyst — monitoring, detecting, and responding to real-time threats. It follows CYVANT's Learn → Practice → Build → Present → Defend → Portfolio → Opportunity approach, moving from foundational concepts through hands-on detection and response work to a capstone AI integration project.

Suggested duration: 8 Weeks (*expanded from the current 6-week estimate to properly cover all 8 modules — flag if you'd rather compress*)

Module 1: Security Foundations — Terminology, OS & Network Basics

- Core cybersecurity terminology and concepts
- Introduction to Windows, Linux, and other operating systems relevant to SOC work
- Networking fundamentals: TCP/IP, subnetting, and common protocols

Module 2: Blue Team Operations & SOC Architecture

- Roles and responsibilities within a Security Operations Center
- Overview of core Blue Team tooling: firewalls, IDS/IPS, SIEM platforms
- Daily SOC operations — monitoring rhythm, maintenance, and how a real shift runs

Module 3: Cyber Threats, Indicators of Compromise & Attack Methodology

- Mapping the current cyber threat landscape
- Understanding and identifying Indicators of Compromise (IoCs)
- Common attack methodologies and how they typically unfold

Module 4: Incidents, Events & Logging

- Distinguishing an incident from an event
- Logging fundamentals
- Log management practices

Module 5: Incident Detection with SIEM

- SIEM fundamentals and its role in security operations
- Setting up and configuring a SIEM system
- Using SIEM for monitoring, detection, and event analysis
- **Hands-on Lab:** configure a SIEM dashboard and triage a simulated alert

Module 6: Strengthening Detection with Threat Intelligence

- Introduction to threat intelligence and why it matters
- Incorporating threat intelligence into daily SOC operations
- Using threat intelligence to sharpen incident detection

Module 7: Incident Response

- Building an incident response plan

- Incident handling in practice
- Post-incident activities and lessons-learned processes
- **Hands-on Lab:** respond to a simulated incident end-to-end, from detection to resolution

Module 8: AI in SOC Analysis – Practical Applications

- AI fundamentals as applied to cybersecurity
- AI-driven threat detection within SIEM workflows
- Automating incident triage with AI tools
- AI-enhanced malware analysis
- Intelligent OSINT and threat intelligence gathering
- **Capstone: AI Integration Project** – apply AI tooling to a realistic SOC workflow challenge

(Per CYVANT's Programme Quality Standard, this capstone should include defined learning outcomes, assessment criteria, a mentor touchpoint, and a public presentation component.)

Career Support

This track includes structured career support layered on top of the technical curriculum:

- **Portfolio building:** students build 3 high-caliber projects across the track, forming the core of their professional portfolio
 - **1 to 1 Mentorship & coaching:** delivered per the standalone **CYVANT Mentorship Support Model** – a field-experienced SOC mentor handles office hours, async support, milestone check-ins, and capstone review (see that document for full structure)
 - **Study group:** peer accountability structure within the cohort's async support channel
 - **Career mapping session:** guidance on how this track connects to specific role targets
 - **Job interview preparation**
 - **Online presence support:** guidance on building LinkedIn/portfolio visibility for job prospects
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Certification Alignment

This track builds toward (external, third-party certifications – confirm exact alignment before publishing):

- CompTIA CySA+
- Blue Team Level 1

- SIEM-platform-specific certifications, depending on which tools the track ultimately standardizes on (e.g. Splunk)

Career Paths This Prepares Students For

- SOC Analyst
- Cybersecurity Analyst
- Security Engineer
- Application Security Engineer
- Cybersecurity Consultant

● Path C: Governance, Risk & Compliance (GRC)

For those drawn to the policy and risk side of security — where technical understanding meets business and regulatory decision-making.

Duration: 16 Weeks

Price: 900,000

Prerequisites: CompTIA Security+ or equivalent skill level

What You'll Learn: Risk Frameworks · Compliance Standards (ISO/NIST) · Policy Writing · Audit Fundamentals

Governance, Risk & Compliance (GRC) — Full Curriculum Breakdown

Tier: 3C — Advanced Specialization **Prerequisites:** CompTIA Security+ **Builds toward (external certifications, not CYVANT credentials):** CRISC (Certified in Risk and Information Systems Control), CISA (Certified Information Systems Auditor), ISO 27001 Lead Implementer/Auditor

Programme Overview

This track is for students drawn to the policy and risk side of security — where technical understanding meets business and regulatory decision-making. It follows CYVANT's Learn → Practice → Build → Present → Defend → Portfolio → Opportunity approach, moving from foundational GRC concepts through hands-on frameworks and tooling to a final portfolio project and job-readiness preparation.

Suggested duration: 9 Weeks (*expanded from the current 4-week estimate to properly cover all 9 modules — flag if you'd rather compress or restructure*)

Module 1: Introduction to GRC & Cybersecurity

- Overview of cybersecurity and GRC as a discipline
- Core concepts: Governance, Risk Management, and Compliance
- Roles and career pathways within GRC

- Getting set up with eramba (open-source GRC platform) for hands-on practice through the rest of the track

Module 2: Regulatory Frameworks & Standards

- Introduction to major frameworks: ISO 27001, NIST, GDPR, PCI-DSS, COBIT, SOC 2
- Understanding a layered/multi-level control framework approach to governance

Module 3: Cybersecurity Risk Management

- Risk management methodology and process
- Risk identification, assessment, evaluation, and response
- Building a risk register and risk matrix
- **Hands-on Lab:** build a sample risk register for a fictional organization

Module 4: Governance Fundamentals

- Writing and reviewing IT policies and procedures
- Document management practices
- Introduction to compliance program structures

Module 5: Third-Party Risk Management (TPRM)

- Introduction to TPRM
- Vendor intake forms and vendor risk assessments
- Due diligence practices
- Full TPRM process walkthrough

Module 6: Audit Management

- Internal audit lifecycle
- Continuous monitoring and metrics
- Reporting compliance status to stakeholders
- **AI-Assisted Learning:** generating a draft internal audit checklist from company policy text

Module 7: Incident Response & Business Continuity (GRC Perspective)

- Incident Response Plans (IRP)
- Business Continuity and Disaster Recovery planning

- **AI-Assisted Learning:** simulating a live incident response scenario through role-play with AI-guided prompts

Module 8: AI in GRC — Practical Applications

- Introduction to AI's role in Governance, Risk & Compliance
- Overview of common AI tools used in compliance work (e.g. ChatGPT, Microsoft Copilot, and dedicated platforms like Diligent and LogicGate)
- Using AI to draft policies, summarize regulations, assess risk, and prepare audit reports
- AI ethics, bias, and responsible automation within compliance work

Module 9: Final Portfolio Project & Career Readiness

- Review and consolidation of GRC competencies covered across the track
- Building a portfolio project that demonstrates real GRC skill (e.g. a full risk assessment, policy set, or audit checklist for a sample organization)
- Preparing for GRC job interviews — mock interviews and CV guidance

(Per CYVANT's Programme Quality Standard, the final project should include defined learning outcomes, assessment criteria, a mentor touchpoint, and a public presentation component.)

Career Support

- **Portfolio building:** the Module 9 final project becomes the core of the student's professional GRC portfolio
 - **Mentorship & coaching:** delivered per the standalone **CYVANT Mentorship Support Model** — a field-experienced GRC mentor handles office hours, async support, milestone check-ins, and final project review (see that document for full structure)
 - **Job interview preparation:** mock interviews and CV guidance, built into Module 9
 - **Study group:** peer accountability within the cohort's async support channel
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Certification Alignment

This track builds toward (external, third-party certifications — confirm exact alignment before publishing):

- CRISC (Certified in Risk and Information Systems Control)
- CISA (Certified Information Systems Auditor)

- ISO 27001 Lead Implementer / Lead Auditor

Career Paths This Prepares Students For

(Inferred from curriculum content – confirm/expand before publishing)

- GRC Analyst
 - Compliance Analyst
 - Risk Analyst
 - IT Auditor
 - Governance Consultant
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Notes

- This replaces the short 4-bullet summary currently in the main course catalog for Path C – once approved, the catalog entry should link to or be replaced by this fuller breakdown.
- This is the third track (after Penetration Testing and SOC Operations) with a dedicated AI-integration module – worth positioning across all three consistently as "every specialization includes practical AI application," which is a genuine differentiator tying back to CYVANT's dual AI + cybersecurity identity.
- Written fresh in CYVANT's own structure and language – safe to publish without copyright concerns, unlike the originally sourced material.